

## Trainings ▾

## Select Service Tools

### Recommended Cummins® Service Tools

- Gear Replacer Kit, Part Number 4918530
- Timing Tool Kit, Part Number 3163021
- Insulated gloves, Part Number 3823730, and/or hot clamp pliers, Part Number 3823732

### With magnetic base drill:

- Magnetic base drill, Part Number 5572871 (60 Hz) or 5572872 (50 Hz)
- Crankshaft adapter and gear remover kit, Part Number 5572864
- Annular cutter bit, Part Number 5572873
- Cutter bit Insert, Part Number 5572875
- Locating gauge, Part Number 4918314
- Gear Remover, Part Number ST647

### Without magnetic base drill:

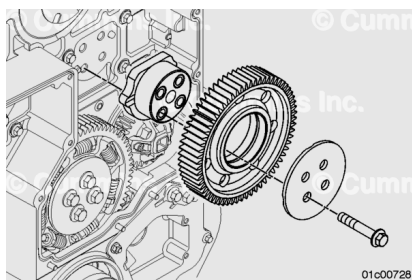
- Drill bit, Part Number 4918535
- Bushing, Part Number 4918538
- Stop collar, Part Number 4918541
- Drill bit, Part Number 4918533
- Drill guide adapter, Part Number 4918276
- Drill bushing, Part Number 4918540
- Drill bit, Part Number 4918537
- Stop collar, Part Number 4918543
- End mill, Part Number 4918532
- Wedge, Part Number 4918546

### Additional Service Items

- Crocus cloth to remove any scratches or nicks

## General Information

Before installation or removal of any gears, you **must** read Engine Base Timing. Refer to Procedure 001-088 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)



## Preparatory Steps

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

### **⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

### **⚠ CAUTION ⚠**

**To reduce the possibility of engine damage, the removal of the lubricating oil pan is required to prevent debris from collecting in the oil pan.**

- Remove the radiator, charge air cooler, and hood, if applicable. See equipment manufacturer service information.
- Disconnect the battery. See equipment manufacturer service information.
- Remove the crankshaft pulley and vibration damper. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-022.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)

- Remove the lower gear cover. Refer to Procedure 001-080 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-080.html)
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-023.html)
- Remove the front gear cover. Refer to Procedure 001-033 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)

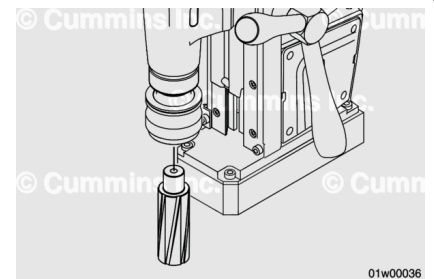
## Remove

With magnetic base drill:

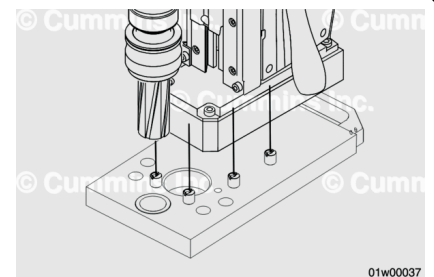
Install annular cutter bit, Part Number 5572873, into the magnetic base drill, Part Number 5572871 (60 Hz) or 5572872 (50 Hz).

Verify the magnetic base drill bit holder is locked into position with the slide lock all the way up.

A click will be felt or heard when the magnetic base drill bit holder locks into place.

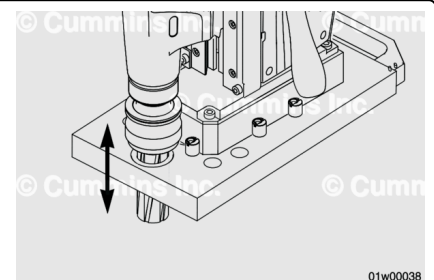


Place the magnetic base drill onto the crankshaft adapter remover plate, Part Number 5572862, close to the bottom and side alignment stop blocks.



Rotate the drill feed handle until the annular cutter bit slides approximately halfway into the drill guide of the crankshaft adapter remover plate.

Verify the annular cutter bit moves freely up and down in the drill guide.



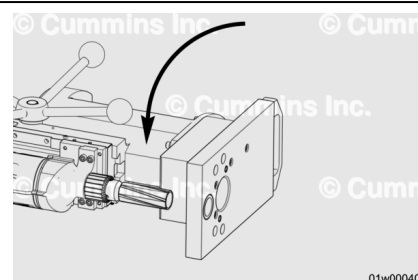
Turn the magnet on.

Verify the annular cutter bit still moves freely up and down in the drill guide.

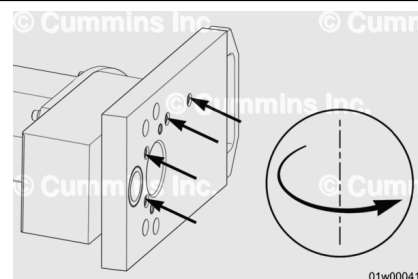
If the annular cutter bit no longer moves freely up and down, turn the magnet off and repeat the previous step.



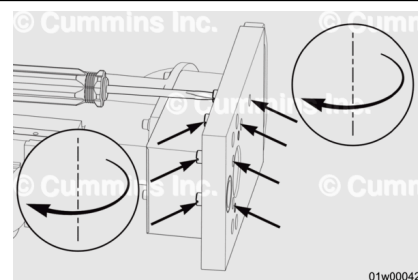
Turn the magnetic base drill and crankshaft adapter remover plate assembly onto its side to expose the alignment stop block fasteners.



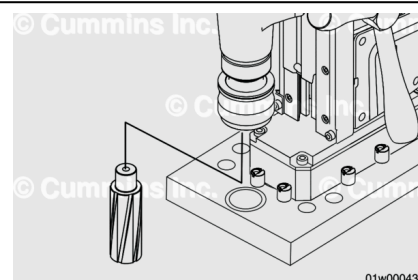
Loosen the fasteners.



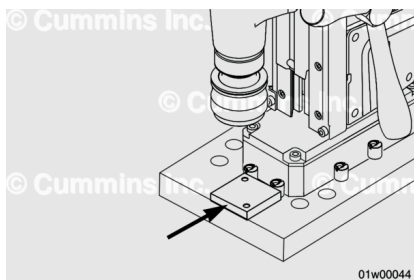
Use a flat tip screwdriver to adjust the alignment stop blocks snug to the drill base, then tighten the fasteners.



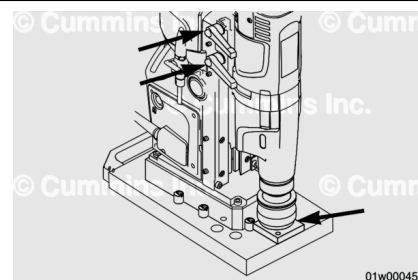
Remove the annular cutter bit from the magnetic base drill bit holder by pulling up on the slide lock to release the bit.



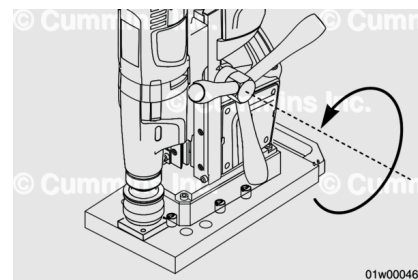
Adjust the magnetic base drill stroke range by placing the gauge plate, Part Number 5572910, over the 1 1/8 in. drill guide on the crankshaft adapter remover plate.



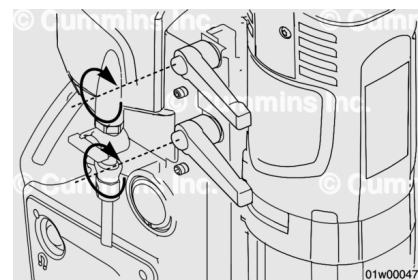
Loosen the magnetic base drill rack fastening levers and lower the drill until the tool holder is resting on the gauge plate.



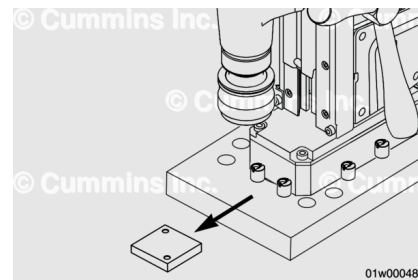
Continue to turn the drill feed handle in the downward direction until the rack bottoms out.



Tighten both fastening levers of the magnetic base drill rack.



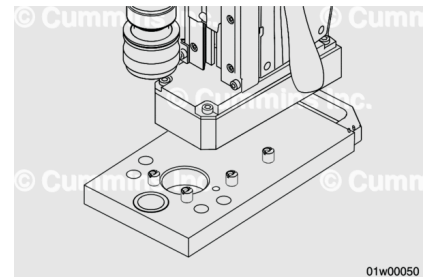
Remove the gauge plate.



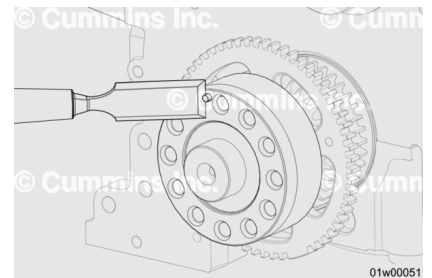
Turn off the magnet of the magnetic base drill.



Remove the magnetic base drill from the crankshaft adapter remover plate.

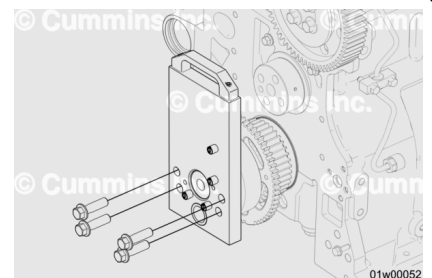


Remove or cut off the dowel pin from the front face of the crankshaft adapter. This will allow the tool to set flat on the crankshaft adapter.

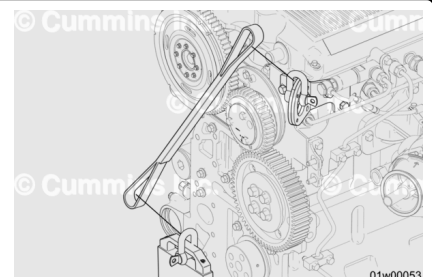


Install the crankshaft adapter remover plate, Part Number 5572862, onto the crankshaft adapter using four of the vibration damper capscrews.

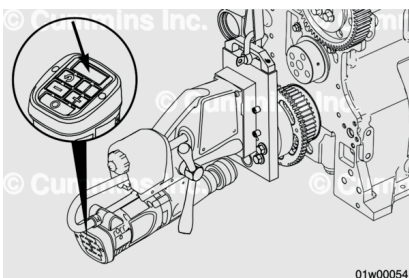
**Torque Value:** 27 n•m [ 20 ft-lb ]



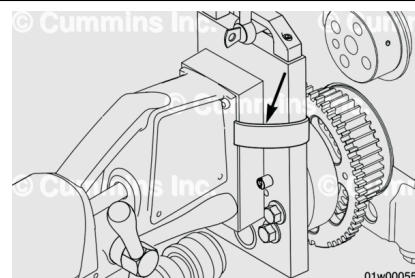
Install the lifting sling, Part Number 5299864, between the adapter remover plate handle and the front lifting bracket with two clevises, Part Number 5299865.



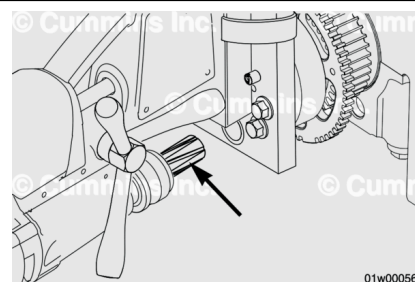
Place the magnetic base drill onto the guide plate against the bottom and side alignment blocks and then turn the magnet on.



Install a safety strap to secure the drill to the plate.

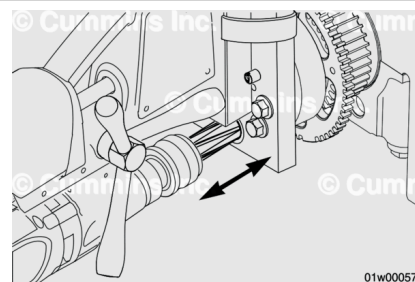


Install annular cutter bit, Part Number 5572873, into the drill tool holder. Verify the tool is locked in position and the slide lock is fully engaged.

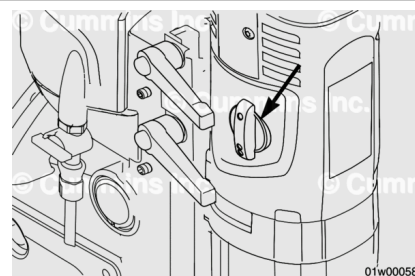


Rotate the drill feed handle until the annular cutter bit slides approximately halfway into the drill guide of the crankshaft adapter remover plate.

Verify the annular cutter bit moves freely in and out of the drill guide.

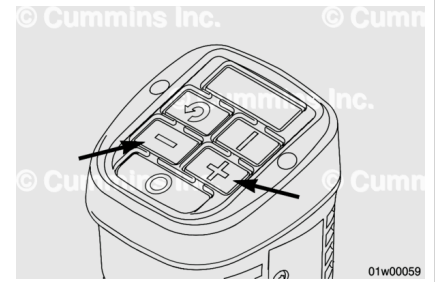


Set the drill gear setting to the slow position one.

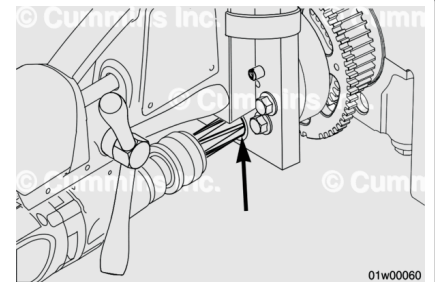


Turn the drill on and adjust speed to approximately 260 rpm.

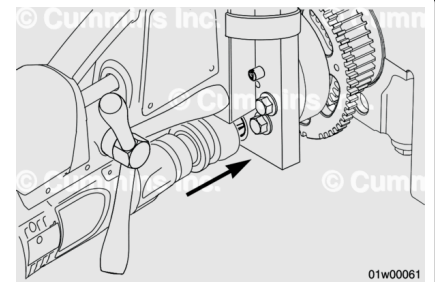
To achieve this, press the minus [ - ] button, repeatedly until the drill reaches the slowest speed, and then press the plus [ + ] button once.



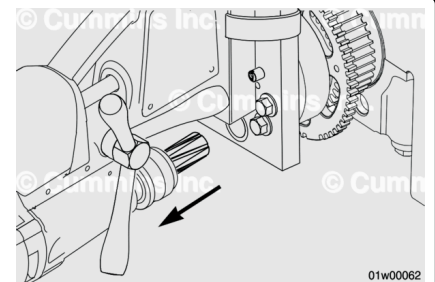
Feed the cutter into the drill guide until it is touching the crankshaft adapter.



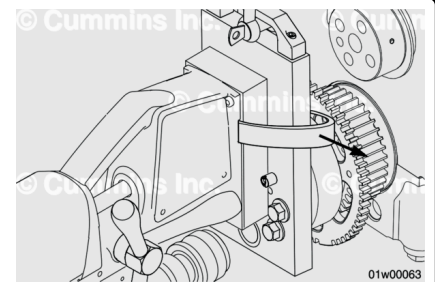
Apply moderate pressure until the drill has reached the end of its travel.



Rotate feed handle in reverse direction to back the bit out of the drilled hole and out of the adapter.



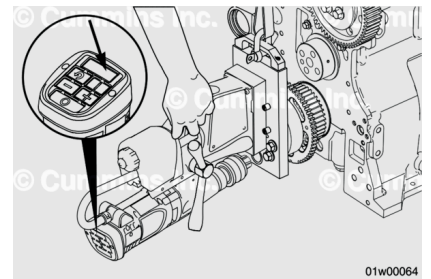
Turn the drill off. Remove the safety strap holding the drill to the adapter remover plate.



Being cautious, hold firmly on to the drill and turn the magnet off. Remove the drill from the adapter remover plate, leaving the lifting sling attached to the crankshaft adapter remover

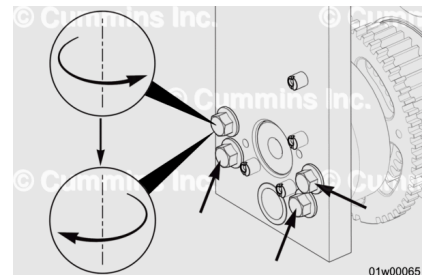


plate.

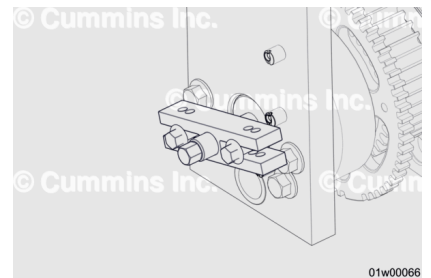


Loosen all four damper capscrews to relax the drilled crankshaft adapter then re-torque.

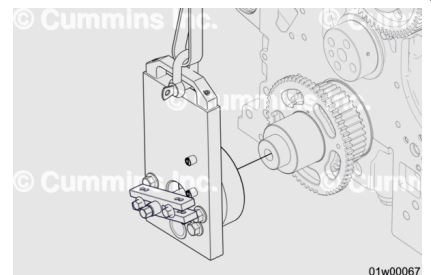
**Torque Value:** 27 n•m [ 20 ft-lb ]



Install the gear remover, Part Number ST-647 or equivalent, onto the crankshaft adapter remover plate using the flange head capscrews, Part Number 5573063, provided in the crankshaft adapter and gear remover kit.



Remove the crankshaft adapter and crankshaft adapter remover plate as an assembly.

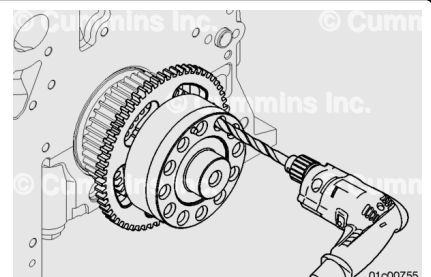


Without magnetic base drill:

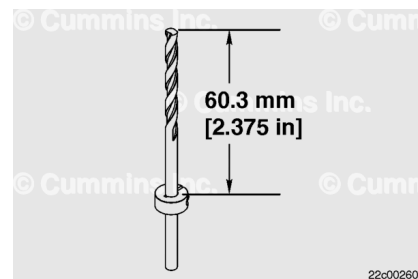
**Note :** If the threaded hole adjacent to the dowel pin is selected, the dowel pin must be removed prior to drilling.

In most instances, this procedure is easiest to perform when starting with a threaded hole near the 12 o'clock position.

Use a 12.7 mm [ 0.50 in ] drill bit, Part Number 4918535, or equivalent, from Gear Replacer Kit, Part Number 4918530, to ream out the threaded hole so the 6.35 mm [ 0.25 in ] bushing, Part Number 4918538, or equivalent, will insert easily.

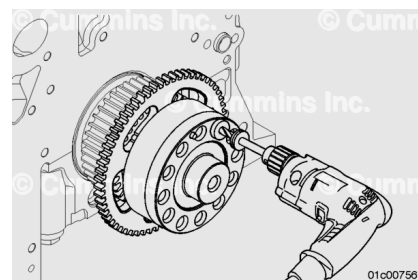


Install the stop collar, Part Number 4918541, or equivalent, from Gear Replacer Kit, Part Number 4918530, onto a 6.35 mm [ 0.25 in ] drill bit, Part Number 4918533, or equivalent, from Gear Replacer Kit, Part Number 4918530, at a depth of 60.3 mm [ 2.375 in ].



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**



**⚠ CAUTION ⚠**

**Do not drill holes completely through the crankshaft adapter or gear. Damage to the engine, crankshaft, and or tools can occur.**

The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

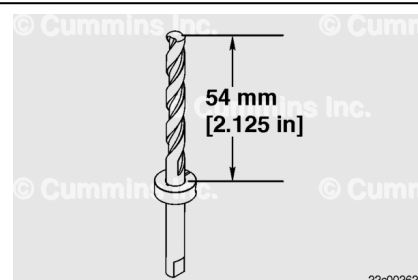
A maximum drill speed of 850 rpm is recommended.

Insert a 6.35 mm [ 0.25 in ] drill bushing, Part Number 4918538, or equivalent, from Gear Replacer Kit, Part Number 4918530, into the reamed hole.

Drill a 6.35 mm [ 0.25 in ] hole to the preset depth.

Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.

Install the stop collar, Part Number 4918542, or equivalent, from Gear Replacer Kit, Part Number 4918530, onto a 12.7 mm [ 0.50 in ] drill bit, Part Number 4918535, or equivalent, at a depth of 54 mm [ 2.125 in ].

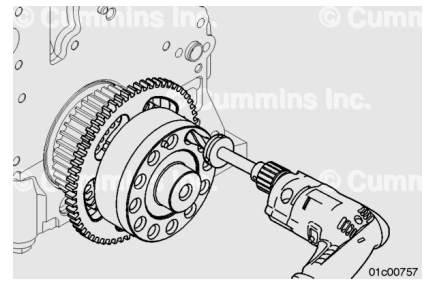


The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

A maximum drill speed of 850 rpm is recommended.

Remove the 6.35 mm [ 0.25 in ] drill bushing, Part Number 4918538, or equivalent, from Gear Replacer Kit, Part Number 4918530. Use the previously drilled 6.35 mm [ 0.25 in ] hole as a pilot, drill a 12.7 mm [ 0.50 in ] hole to the preset depth.

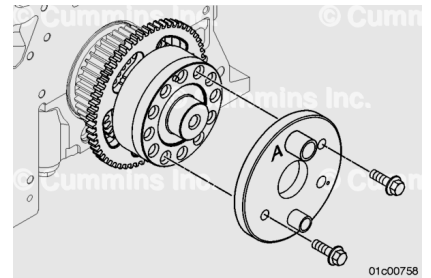
Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.



Install the drill guide adapter, Part Number 4918276, from Gear Replacer Kit, Part Number 4918530, onto the crankshaft nose so the drill guide bushing labeled "A" is aligned with the previously drilled 12.7 mm [ 0.50 in ] hole.

Install and tighten two existing damper capscrews through the drill guide adapter.

**Torque Value:** 13.6 n·m [ 120 in-lb ]



The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

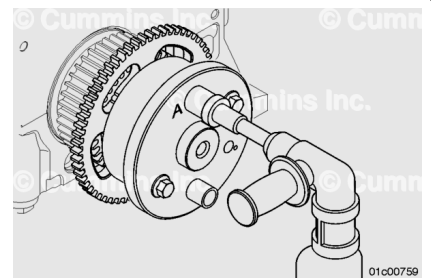
A maximum drill speed of 850 rpm is recommended.

Install the 19.05 mm [ 0.75 in ] drill bushing, Part Number 4918540, or equivalent, from Gear Replacer Kit, Part Number 4918530, into the drill guide bushing "A".

Drill a 19.05 mm [ 0.75 in ] hole. Use drill bit, Part Number 4918536, or equivalent, from Gear Replacer Kit, Part Number 4918530, to a depth of the previous 12.7 mm [ 0.50 in ] hole.

Remove the 19.05 mm [ 0.75 in ] drill bit.

Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.

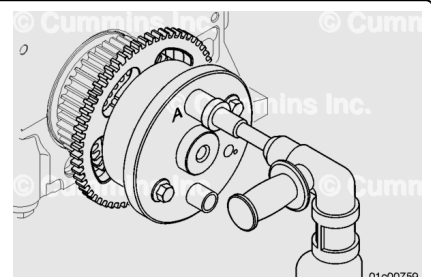


The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

A right angle, clutched, 1/2 inch drive drill at a maximum drill speed of 300 rpm is recommended.

Drill a 25.4 mm [ 1.00 in ] hole. Use drill bit, Part Number 4918537, or equivalent, from Gear Replacer Kit, Part Number 4918530, to a depth of the previous 19.05 mm [ 0.75 in ] hole.

Remove the 25.4 mm [ 1.00 in ] drill bit.

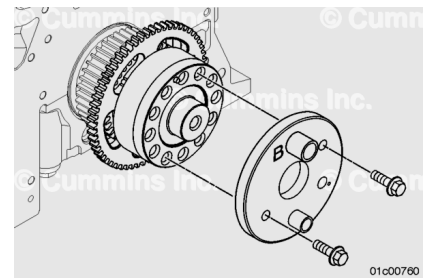


Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.

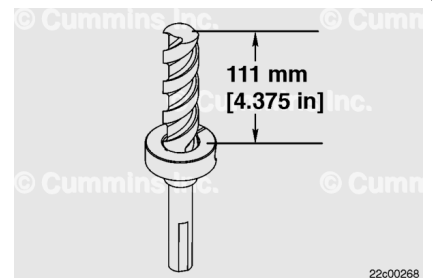
Remove the drill guide adapter from the crankshaft and rotate the drill guide adapter 180 degrees so the drill guide bushing labeled "B" is aligned with the previously drilled hole.

Install and tighten two existing damper capscrews through the drill guide adapter.

**Torque Value:** 13.6 n•m [ 120 in-lb ]



Install the stop collar, Part Number 4918543, or equivalent, onto a 19.05 mm [ 0.75 in ] end mill, Part Number 4918532, both from Gear Replacer Kit, Part Number 4918530, at a depth of 111 mm [ 4.370 in ].



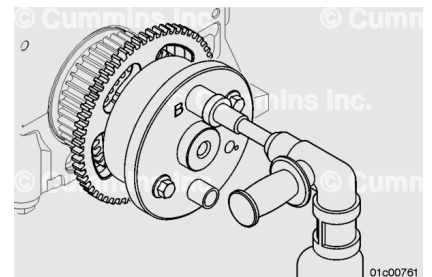
The use of a generous amount of cutting fluid will increase tool life and ease the drilling operation.

A right angle, clutched, 1/2 inch drive drill at a maximum drill speed of 300 rpm is recommended.

Bore a 19.05 mm [ 0.75 in ] hole to the preset depth.

Remove the 19.05 mm [ 0.75 in ] end mill, the two damper capscrews, and the drill guide adapter.

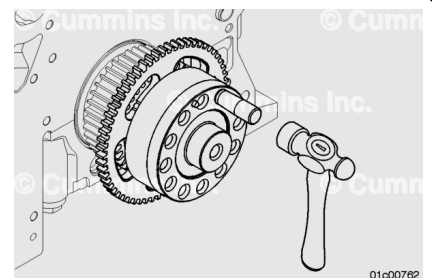
Clean out the debris. Use low pressure air and a shop towel to cover the hole as necessary.



Insert the wedge, Part Number 4918546, from Gear Replacer Kit, Part Number 4918530, into the 19.05 mm [ 0.75 in ] drilled hole.

Strike the wedge firmly with a hammer until the crankshaft adapter splits.

The crankshaft adapter will be easily removable by hand after splitting, however, a suitable puller can be used to facilitate the removal.

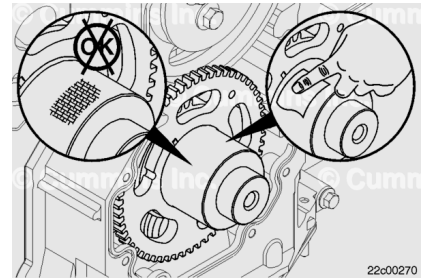


## Clean and Inspect for Reuse

Inspect the crankshaft, tone wheel, and crankshaft gear for damage.

Inspect the machined surface for scratches or nicks.

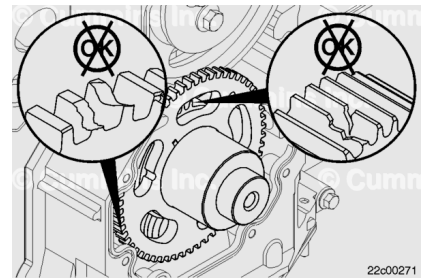
Use a crocus cloth to remove any scratches or nicks. If scratches or nicks can be felt with a fingernail after the crankshaft has been polished with the crocus cloth, the crankshaft **must** be replaced.



Inspect the crankshaft gear and tone wheel for excessive wear or damage.

If any teeth on the crankshaft gear are missing, chipped, or pitted, the crankshaft gear **must** be replaced.

If any teeth on the tone wheel are missing, the tone wheel **must** be replaced.



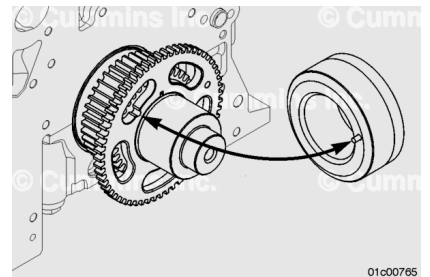
## Install

### **⚠ WARNING ⚠**

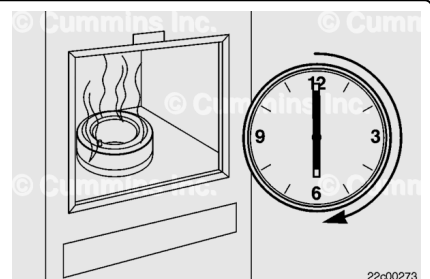
To reduce the possibility of personal injury, wear protective gloves when handling parts that have been heated.

**Note :** During the installation of the crankshaft adapter, the pin located on the back side of the crankshaft adapter **must** align with the slot on the front surface of the tone wheel.

Do **not** lubricate the crankshaft prior to installing the crankshaft adapter.



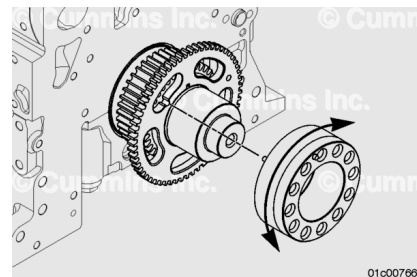
Heat the crankshaft adapter for 6 hours at 260°C [ 500°F ] before attempting the installation.



**Note :** The crankshaft adapter must be installed within 15 seconds after being removed from the oven.

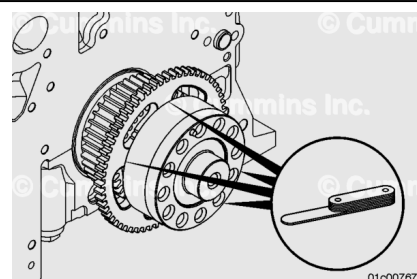
Install the crankshaft adapter. Use a twisting motion until the pin is aligned in the slot on the tone wheel.

Hold the crankshaft adapter against the tone wheel for approximately 30 seconds.



To check the installation, attempt to insert a 0.25 mm [ 0.010 in ] feeler gauge between the crankshaft adapter and the tone wheel at four locations approximately 90 degrees apart.

If the feeler gauge can be inserted, the crankshaft adapter **must** be removed and replaced.



## Finishing Steps

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Install the front gear housing and gears. Refer to Procedure 001-033 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-033.html)
- Install the front crankshaft seal. Refer to Procedure 001-023 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-023.html)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
- Install the lower gear cover. Refer to Procedure 001-080 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-080.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7  
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html).
- Statically time the engine. Refer to Procedure 001-088 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

**Note :** If the timing pin, Part Number 3163020, from Timing Tool Kit, Part Number 3163021, can not be inserted when the damper marking "Insert Pin" is aligned with the marking on the lower gear, the tone wheel or crankshaft adapter is not properly aligned with the crankshaft.

- Install the crankshaft pulley and vibration damper. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-022.html>)
- Connect the batteries. See equipment manufacturer service information.
- Install the radiator, charge air cooler, and hood, if applicable. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks.

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**Last Modified: 11-Aug-2022**

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